

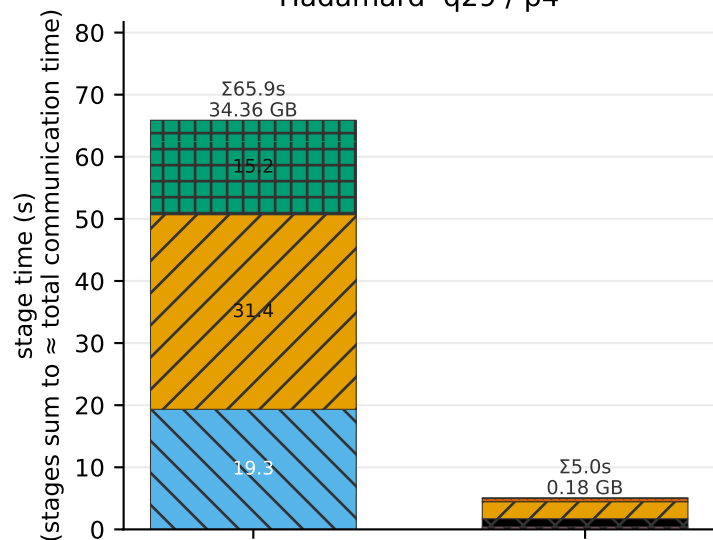
Inside the communication time, compression OFF vs ON: which stage the seconds go to QuEST GPU+MPI, 29 qubits / 4 ranks, 4xRTX 2080 Ti (landonia01); slowest of the 4 ranks shown — it sets the wall time

Compression ON replaces most raw-payload MPI time with smaller GPU compression stages; the payload (GB) is shown above each bar.

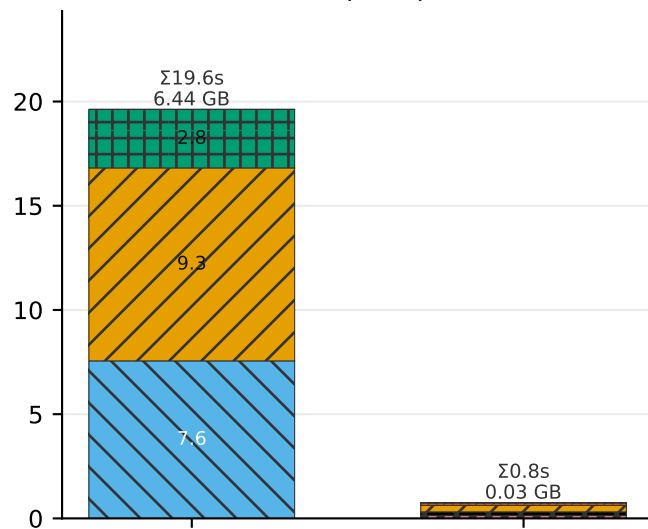
Top: seconds and stage total Σ . Bottom: the same communication stages normalized to 100%. Stages below 2% are folded into 'other copies'.

Computation, start-up, and execution overhead are outside this communication-only view.

Hadamard q29 / p4



QFT q29 / p4



Random q29 / p4

